Classical Field Theory — Exercise Sheet 26

Scattering evolution and soliton shifts

Hints and answers

Use the standing real KdV solution $u \in C^\infty(\mathbb{R}; \mathcal{S})$ of Lecture 24 and the Jost/scattering conventions of Lecture 25. The first problem fills the analytic step left sketched in Lecture 26. The later finite-rank construction prescribes its norming coefficients; it makes no additional theorem about birth of eigenvalues.

Exercise 26.1 (Time differentiation at real energy). Fix real $k \neq 0$. For the right Jost solution put $f_+(t,x,k) = e^{\mathrm{i}kx} m(t,x,k)$, with

$$m = 1 + \int_x^\infty \frac{e^{2\mathrm{i}k(y-x)} - 1}{2\mathrm{i}k}\, u(t,y) m(t,y,k)\,\mathrm{d}y.$$

(a) Prove that this Volterra construction is continuously differentiable in t. On every compact time interval prove $m_t \to 0$ as $x \to +\infty$, together with the spatial derivative limits needed to compare both the function and derivative tails of $(f_+)_t - Bf_+$. Here $B = -4\partial_x^3 + 6u\partial_x + 3u_x$.

(b) Prove the right and left temporal equations, justify differentiability of a, b from finite-point Wronskians, and prove $r(k,t) = r(k,t_0)e^{8\mathrm{i}k^3(t-t_0)}$. Do not pair the oscillatory Jost solutions in L^2.

Hint. Differentiate each ordered Volterra product once in time. The sum over its possible differentiated factors has a factorial majorant. Obtain higher spatial derivatives from the modified scalar ODE. Jost uniqueness requires both a function tail and a first-derivative tail.

Answer. At real energy,

$$(f_+)_t = Bf_+ - 4\mathrm{i}k^3 f_+, \qquad (f_-)_t = Bf_- + 4\mathrm{i}k^3 f_-,$$
$$a_t = 0, \qquad b_t = 8\mathrm{i}k^3 b.$$

The required tail estimates follow from compact-time L^1 tails of u, u_t and their spatial derivatives.

Exercise 26.2 (A positive finite-rank inverse equation). Take distinct $\kappa_j > 0$ and $c_j > 0$, $1 \le j \le N$, with $F(s) = \sum_j c_j e^{-\kappa_j s}$. Consider the right Marchenko equation

$$K(x,y) + F(x+y) + \int_x^\infty K(x,z)F(z+y)\,\mathrm{d}z = 0, \qquad y \ge x.$$

The finite positive reflectionless case of the inverse-scattering theorem is a *given*: these data are admissible and reconstruct a real rapidly decaying potential by $u = -2\,\mathrm{d}K(x,x)/\mathrm{d}x$. It is the zero-background specialization of the cited theorem in Lectures 25–26: Egorova–Grunert–Teschl, *Nonlinearity* **22** (2009), §4, Theorems 4.3–4.4, equations (4.26)–(4.30). Their $(\gamma_j^+)^2$ is c_j.

(a) Reduce the equation to a real symmetric $N \times N$ linear system. Prove that its matrix is positive definite and invertible for every real x.

(b) With $v_j(x) = \sqrt{c_j} e^{-\kappa_j x}$, define $\mathsf{C}_{jl} = v_j v_l/(\kappa_j + \kappa_l)$ and $\tau = \det(I + \mathsf{C})$. Prove $K(x,x) = (\log\tau)'$ and $u = -2(\log\tau)''$. Explain why no real pole arises from these positive data.

Hint. Write $K = -\sum_j z_j(x)e^{-\kappa_j y}$ and divide its coefficient equations by $\sqrt{c_j}$. The matrix C is an integral of a real outer product on $[x,\infty)$. Use the determinant derivative formula.

Answer. Putting $q_j = z_j/\sqrt{c_j}$ gives $(I+\mathsf{C})q = v$ and $K(x,x) = -v^{\mathsf{T}}q$. Since $\mathsf{C}' = -vv^{\mathsf{T}}$, $\tau'/\tau = -v^{\mathsf{T}}(I+\mathsf{C})^{-1}v = K(x,x)$. The matrix $I+\mathsf{C}$ is positive definite, so $\tau > 0$.

Exercise 26.3 (Two pulses and their shifts). Take $\kappa_1 > \kappa_2 > 0$ and prescribe $c_j(t) = c_j(0)e^{8\kappa_j^3 t}$, with $c_j(0) > 0$. Put

$$x_{j,0} = \frac{1}{2\kappa_j}\log\frac{c_j(0)}{2\kappa_j}, \qquad \eta_j = -2\kappa_j(x - 4\kappa_j^2 t - x_{j,0}).$$

(a) Evaluate the two-state determinant and find its interaction coefficient d_{12} in $\tau = 1 + e^{\eta_1} + e^{\eta_2} + d_{12}e^{\eta_1+\eta_2}$.

(b) Verify directly that $u = -2\partial_x^2 \log\tau$ solves KdV. One may use, after deriving it by differentiating $\log\tau$,

$$\tau\tau_{xt} - \tau_x\tau_t + \tau\tau_{xxxx} - 4\tau_x\tau_{xxx} + 3\tau_{xx}^2 = 0$$

as a sufficient condition.

(c) Find the incoming and outgoing centre lines of each separated pulse and its net shift. Justify the two spatial derivatives in each moving-frame limit, rather than compare only pointwise values of τ.

The parameters $x_{j,0}$ describe the norming coefficients. They need not be actual minima positions during the interaction.

Hint. For the PDE check expand the symmetric bilinear expression on pairs of exponentials; only the mixed coefficient is nontrivial. For the shifts, factor out the large exponential in each moving frame. Its logarithm is affine in x and contributes nothing to the second derivative.

Answer. The coefficient and shifts are

$$d_{12} = \left(\frac{\kappa_1 - \kappa_2}{\kappa_1 + \kappa_2}\right)^2,$$

$$\Delta x_1 = -\frac{\log d_{12}}{2\kappa_1} > 0, \qquad \Delta x_2 = \frac{\log d_{12}}{2\kappa_2} < 0.$$

The faster pulse advances and the slower one is delayed.